

B.Tech. Degree IV Semester Examination April 2014**ME 1404 APPLIED THERMODYNAMICS**
(2012 Scheme)

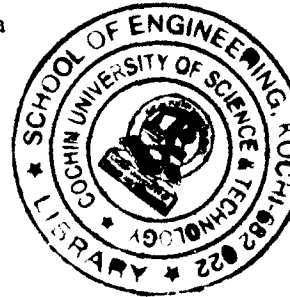
Time : 3 Hours

Maximum Marks : 100

PART A(Answer *ALL* questions)

(8 x 5 = 40)

- I.
 - (a) What are Helmholtz function and Gibbs function? How are these functions related to availability analysis?
 - (b) Explain the first law of thermodynamics for closed system and for a cyclic process. What are the limitations of first law?
 - (c) Explain the process of formation of steam at 1atm from -10°C to 260°C using a T-S diagram.
 - (d) Write a note on binary vapour cycle, with neat sketches.
 - (e) Why are supersonic nozzles convergent-divergent?
 - (f) Describe velocity compounding of impulse turbines.
 - (g) Explain Dalton's law of partial pressure.
 - (h) Adiabatic mixing of two fluids are irreversible. Justify.

**PART B**

(4 x 15 = 60)

- II.
 - (a) Derive the Maxwell equations from fundamentals. (10)
 - (b) Derive the first and second Tds equation. (5)
- OR**
- III.
 - (a) Show that the inequality of Clausius is valid for all possible cycles. (7)
 - (b) Define and explain Carnot theorem. Show that the efficiency of a reversible engine is independent of the nature or amount of the working substance undergoing the cycle. (8)
- IV.

In a single heater regenerative cycle the steam enters the turbine at 30 bar, 400°C and the exhaust pressure is 0.1 bar. The feed water heater is a direct contact type which operates at 5 bar. Find (i) The efficiency and steam rate of the cycle (ii) The increase in mean temperature of heat addition, efficiency and steam rate as compared to the Rankine cycle (without regeneration). Neglect pump work. (15)
- OR**
- V.
 - (a) A boiler generates 45×10^3 kg/hr of wet steam at 10 bar pressure and .95 dryness fraction. The boiler receives water at 40°C and the local coal burns in the furnace at the rate of 5×10^3 kg/hr. Calculate the equivalent evaporation of boiler per kg of coal burnt. (8)
 - (b) Explain the working Lamont boiler, with a neat sketch. (7)

(P.T.O.)

- VI. Steam at a pressure of 7 bar and 180°C is discharged through a convergent-divergent nozzle at 1 bar. The mass flow rate of steam is estimated to 2 kg/s. The pressure at the nozzle throat is 4 bar and the steam enters the nozzle with velocity of 75 m/s expansion upto nozzle throat is isentropic and the frictional resistance equivalent to 60 kJ/kg of steam reheats the steam in the divergent section. Make calculations for: (15)
- Suitable areas for the throat and exit
 - Overall efficiency of nozzle based on enthalpy drop between the actual state of steam and the back pressure.

OR

- VII. The velocity of steam at inlet to a single row impulse turbine is 400m/s and the nozzle angle is 20° . The mean blade speed is 150 m/s and the axial thrust on the blade is estimated to be zero. Make calculations for: (15)
- Inlet and outlet angles of moving blades
 - Power developed for a steam flow rate of 1.5kg/s
 - Magnitude and direction of velocity of steam at exit
- Neglect the effect of friction when passing through blade passages and obtain your solution from the geometry of velocity diagrams.

- VIII. Two vessels are connected by a valve. One vessel contains 0.6kg mole of nitrogen gas at 50°C and 15 bar pressure. The other vessel holds 3.5kg of the same gas at 25°C and 6.5 bar. The valve is opened and the gases are allowed to mix while having heat interaction with the surroundings. If the final equilibrium temperature is 30°C , (15)
- Calculate the final equilibrium pressure and the amount of heat transferred to the surroundings
 - Considering the vessels to be perfectly insulated, find the final temperature and pressure, which would have been reached. Take adiabatic exponent $\gamma = 1.4$.

OR

- IX. A mixture of ideal gases consist of 3kg of nitrogen and 5kg of carbon dioxide at a pressure of 300kPa and a temperature of 20°C . Find (i) The mole fraction of each constituent (ii) The equivalent molecular weight of the mixture (iii) The equivalent gas constant of the mixture (iv) The partial pressures and the partial volumes (v) The volume and density of the mixture (vi) The C_p and C_v of the mixture. (15)
